The exponential function and continuous models

Lecture and worked examples

Chapter path

Objectives. Use the algebraic properties; differentiate; study growth or decay; link the exponential to geometric sequences.

Prerequisites. Powers, geometric sequences, differentiation and sign.

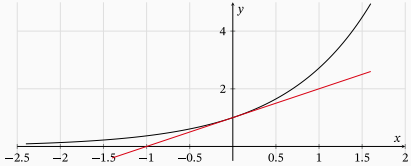
Yearly reference point. 2-11 February: 8 hours

Defining a new function

The exponential function is the unique function f differentiable on $\mathbb{R}$ satisfying $f' = f$ and $f(0) = 1$. Its existence and uniqueness are accepted. It is written exp, then e^x with $e = \exp(1) \approx 2.71828$. The number e is a constant; x is the variable. The exponential is strictly positive.

Defining a new function

$\exp(0) = 1$, $\exp(1) = e$, $\exp\prime(0) = 1$. The tangent at 0 is therefore $y = 1 + x$. The notation e^x extends the properties of powers to every real exponent.



Defining a new function

$e^0 = 1$, not 0. The function never takes the value 0, even for negative x.

Defining a new function

What is the equation of the tangent to e^x at 1?

Algebraic properties and comparisons

For all real x,y, $e^{x+y} = e^x e^y$, $e^{-x} = \frac{1}{e^x}$, $e^{x-y} = \frac{e^x}{e^y}$. For integer n, $(e^x)^n = e^{nx}$. The function is strictly increasing on $\mathbb{R}$, so $e^u = e^v \iff u = v$ and $e^u < e^v \iff u < v$.

Algebraic properties and comparisons

$e^{2x}e^{1-x} = e^{x+1}$. $e^{3x-2} = e^{x+4}$ is equivalent to $3x - 2 = x + 4$, so x=3. $e^{2x} \leq 1 = e^0$ is equivalent to $x \leq 0$.

Algebraic properties and comparisons

e^{x+y} is a product, not in general $e^x + e^y$. We do not solve $e^x = 2$ using a logarithm, which is not studied in Première.

Algebraic properties and comparisons

Simplify $\dfrac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

Differentiating and studying

For real a, $t \mapsto e^{at}$ is differentiable and its derivative is ae^{at}. For $e^{at+b} = e^b e^{at}$, the derivative is ae^{at+b}. The product and quotient rules apply with the derivatives already known. The sign of the exponential often simplifies the study of the sign of the derivative.

Differentiating and studying

For $f(x) = (x+1)e^{-x}$, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. Since $e^{-x} > 0$, the derivative has the sign of $-x$: global maximum 1 at 0.

Differentiating and studying

Do not forget the factor a or the negative sign in e^{-x}. The exponential factor is positive, not necessarily increasing.

Differentiating and studying

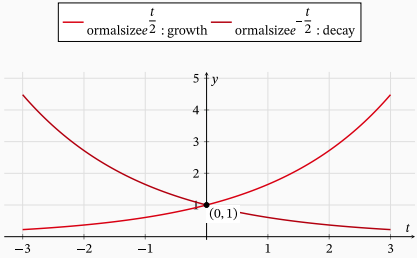
Differentiate $g(t) = 3e^{-2t}$ and study its monotonicity.

From discrete to continuous

A model $Q(t) = Q_0 e^{kt}$, with $Q_0 > 0$, represents growth if $k > 0$, decay if $k < 0$, constancy if $k = 0$. For a fixed step h, $\dfrac{Q(t+h)}{Q(t)} = e^{kh}$. The values at integer times form a geometric sequence with common ratio e^k. The unit of k is the inverse of the unit of time.

From discrete to continuous

$Q(t) = 100e^{-0.2t}$, t in hours: $Q(0) = 100$, $\dfrac{Q(t+1)}{Q(t)} = e^{-0.2}$. A simulation can look for the first whole number of hours at which Q falls below 50; the exact equation is not solved with a logarithm in this chapter.



From discrete to continuous

The relative rate per discrete unit of time is $e^k - 1$, not exactly k; they are close only for small k.

From discrete to continuous

For $Q(t) = 5e^{0.3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$.

Proving an inequality

The inequality $e^x \geq 1 + x$, true for every real x, is obtained by studying the function $g(x) = e^x - 1 - x$: the curve of the exponential lies above its tangent at 0.

Proving an inequality

With $x = 0.1$, the inequality gives $e^{0.1} \geq 1.1$; the calculator displays $e^{0.1} \approx 1.10517$. With $x = -1$, it gives $e^{-1} \geq 0$, which is true but not very precise.

Proving an inequality

A graph showing a curve above a line does not prove the inequality for all real numbers.

Proving an inequality

Show that $e^{2x} \geq 1 + 2x$ for every real number x.